

L13a: Synthetic Markets with Jump Hidden Markov Models

CHEME 5660 Cornell University

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Objectives for Today

Today, we build synthetic markets: simulated price histories with the features of real ones. How would a trading method do on thousands of plausible paths?

Today's concepts

- Markov chains and hidden Markov models: states, transition matrices, stationary distributions, emissions.
- A jump hidden Markov model of one asset: binned states, counted transitions, Student-t emissions, jump episodes.
- A synthetic market of many stocks: shared market and sector draws plus each stock's own draw.

What You Need to Know

Core concepts

- A Markov chain is a set of states $\mathcal{S}$ and a transition matrix $\mathbf{P}$ whose rows sum to one; its stationary distribution satisfies $\bar{\pi} = \bar{\pi}\mathbf{P}$.
- The jump hidden Markov model bins the daily growth rate into states, estimates $\mathbf{P}$ by counting, and draws each day from a Student-t of the day's state. Jump episodes (probability ϵ, mean length λ) keep volatility clustering.
- A synthetic stock is $g_i = \alpha_i + \beta_i\, g_m + \delta_i\, f_s + \psi_i\, \tilde{\varepsilon}_i$: shared market and sector draws make stocks move together.

For understanding (you won't reproduce these)

- Power iteration, the tuning objective for ϵ and λ, and how the sector loadings and own-draw scale are chosen.

Lectures and Examples

Lecture:

CHEME-5660-L13a-Lecture-JumpHMM-SyntheticMarkets-Fall-2026.ipynb

Examples:

CHEME-5660-L13a-Example-JumpHMM-SPY-Fall-2026.ipynb

CHEME-5660-L13a-Example-JumpHMM-SectorFactor-Fall-2026.ipynb

Why Synthetic Markets?

A backtest runs a method on one historical path, and a method that did well may have been lucky. A synthetic market generates many plausible paths. For stress tests, its paths need:

- Fat tails: large daily moves are far more common than a normal distribution predicts.
- Volatility clustering: large moves follow large moves, so the size of the daily move stays correlated for weeks.
- Co-movement: stocks move together, through the market and their sectors.

Geometric Brownian motion can have co-movement through correlated shocks, but neither fat tails nor volatility clustering.

Markov Chains

A discrete Markov model $\mathcal{M} = (\mathcal{S}, \mathbf{P})$ has a state space $\mathcal{S}$, the set of values the system can take, and a transition matrix $\mathbf{P}$. It has the Markov property: given the present state, the past does not matter:

$$P(X_{n+1} = s_{n+1} \mid X_0 = s_0, \ldots, X_n = s_n) = P(X_{n+1} = s_{n+1} \mid X_n = s_n)$$

The one-step transition probability from s_i to s_j is $p_{ij} = P(X_{n+1} = s_j \mid X_n = s_i)$, the same for every n (time-homogeneous).

The Transition Matrix

The p_{ij} fill the transition matrix $\mathbf{P} \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{S}|}$: rows are current states, columns next states, $p_{ij} \geq 0$, and each row sums to one. For weather with $\mathcal{S} = \{\text{Sunny}, \text{Rainy}\}$:

$$\mathbf{P} = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix} \qquad \pi_1 = \pi_0 \mathbf{P} = [1,\, 0] \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix} = [0.8,\, 0.2]$$

If today is Sunny, tomorrow is Sunny with probability 0.8. The distribution π is a row vector: $\pi'_j = \sum_i \pi_i p_{ij}$.

Estimating the Transition Matrix

We do not know $\mathbf{P}$; we estimate it from observed states $X_1, \ldots, X_T$. With n_{ij} the number of days in s_i followed by a day in s_j:

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_{k=1}^{|\mathcal{S}|} n_{ik}}$$

the share of days in s_i that moved to s_j. Each row sums to one by construction; this is the maximum likelihood estimate, and how a market generator estimates its transition matrix.

The Stationary Distribution

The stationary distribution $\bar{\pi}$ is unchanged by a step, $\bar{\pi} = \bar{\pi}\mathbf{P}$: the long-run share of time in each state, and where a generator starts its paths. Power iteration applies $\mathbf{P}$ until nothing changes:

- Start from the uniform $\pi = [1/N, \ldots, 1/N]$ over the N states.
- Repeat $\pi' \leftarrow \pi\mathbf{P}$ until $\|\pi' - \pi\|_1 \leq$ tol or T iterations; return $\bar{\pi} \leftarrow \pi'$.

This needs an ergodic $\mathbf{P}$; software often solves $\bar{\pi} = \bar{\pi}\mathbf{P}$ directly instead.

Hidden Markov Models

In a hidden Markov model $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{P}, \mathbf{E})$ we do not see the states, only what they emit: observations $\mathcal{O}$ with emission probabilities $e_{ij} = P(O_n = o_j \mid X_n = s_i)$, rows summing to one. To sample: draw $X_0 \sim \bar{\pi}$; then each step draws X_n from row X_{n-1} of $\mathbf{P}$ and O_n from row X_n of $\mathbf{E}$.

Continuous emissions: a growth rate is a number, so each state has an emission *distribution* instead of a row of $\mathbf{E}$: $g_n \mid X_n = s_k \sim$ emission distribution of s_k.

The Jump Hidden Markov Model

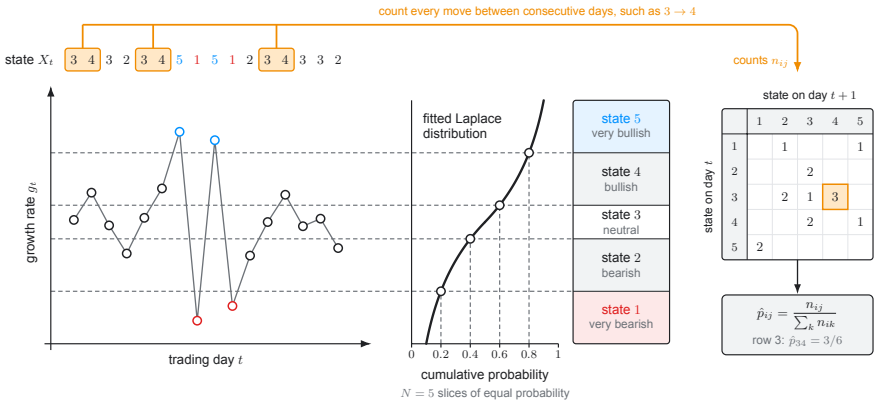
For the daily growth rate $g_t = \frac{1}{\Delta t}\ln(S_t/S_{t-1})$ (1/yr) of one asset:

- States: fit a Laplace distribution to the growth rates and cut it into N bins of equal probability; state 1 holds the most negative growth rates.
- Transitions: assign each day to its bin and count, $\hat{p}_{ij}$.
- Emissions: in state s_k, $g_t \mid X_t = s_k \sim \mu_k + \sigma_k T_\nu$: μ_k, σ_k the mean and standard deviation of the days in that state, T_ν a Student-t with ν fixed.

In the data the states are observed bins; in simulation they are hidden, and only the growth rates are reported.

Fitting the Model

Each state is a slice of the fitted Laplace distribution and a mood of the market. We count the moves between states:



Algorithm: Fit the Model

Given the daily growth rates $g_1, \ldots, g_T$ of one ticker, the number of states N, and ν:

1. Laplace: $\mu \leftarrow$ the median of the growth rates, and $b \leftarrow \frac{1}{T}\sum_t |g_t - \mu|$.
2. States: $X_t \leftarrow s_k$, where $(k-1)/N \le F_L(g_t) < k/N$.
3. Count: $n_{ij} \leftarrow n_{ij} + 1$ for each move from $X_{t-1} = s_i$ to $X_t = s_j$.
4. Transitions: $\hat{p}_{ij} \leftarrow n_{ij} / \sum_k n_{ik}$.
5. Emissions: μ_k, $\sigma_k \leftarrow$ the mean and standard deviation of the growth rates of the days in state s_k.
6. Start: compute the stationary distribution $\bar{\pi}$ of $\hat{\mathbf{P}}$.

Full version: the lecture notebook

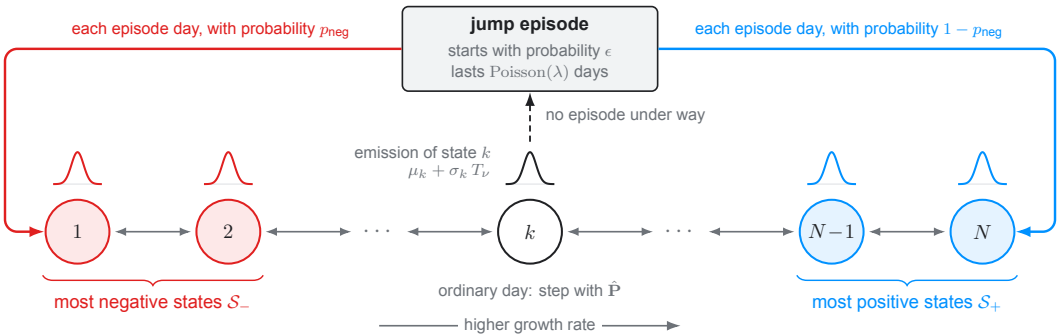
Why Add Jumps?

The counted chain forgets quickly: after a volatile day, a few transitions usually return it to the middle states, so $|g_t|$ is correlated for only a few days. Jump episodes keep volatile periods going:

- With no episode under way, an episode starts with probability ϵ.
- It lasts a Poisson number of days with mean λ.
- Each episode day is drawn from the few most extreme states, negative or positive: a run of large moves of both signs.

The Jump Hidden Markov Model: One Day

An ordinary day steps with $\hat{\mathbf{P}}$. An episode day draws its state from the most negative or the most positive states:



Algorithm: Sample a Path

Draw $X_1 \sim \bar{\pi}$ and g_1 from its emission, and set $\texttt{remaining} \leftarrow 0$. Each day $t = 2, \ldots, T$ runs in this order:

1. New episode: if $\texttt{remaining} = 0$, draw $u \sim \text{Uniform}(0, 1)$. If $u < \epsilon$, draw $\texttt{remaining} \sim \text{Poisson}(\lambda)$.
2. Episode day: if $\texttt{remaining} > 0$, draw X_t uniformly from the most negative states $\mathcal{S}_-$ with probability p_{neg}, otherwise from the most positive states $\mathcal{S}_+$. Set $\texttt{remaining} \leftarrow \texttt{remaining} - 1$.
3. Ordinary day: if $\texttt{remaining} = 0$, draw X_t from row X_{t-1} of $\hat{\mathbf{P}}$.
4. Emission: $g_t \leftarrow \mu_k + \sigma_k T_\nu$, where $X_t = s_k$.

Full version: the lecture notebook

Tuning and Checking the Model

A grid search picks (ϵ, λ). On the M simulated paths with an episode, it scores each path's autocorrelation $\hat{\rho}_{m,k}$ of $|g_t|$ and kurtosis $\hat{\kappa}_m$ against the data's ρ_k, κ:

$$J(\epsilon, \lambda) = \frac{1}{M} \sum_{m=1}^{M} \Big[\sum_{k=1}^{25} \left(\rho_k - \hat{\rho}_{m,k}\right)^2 + 0.2 \left(\kappa - \hat{\kappa}_m\right)^2 \Big]$$

The score is noisy, so the chosen pair can change with the seed.

Example: CHEME-5660-L13a-Example-JumpHMM-SPY-Fall-2026.ipynb

Many Assets: The Sector Factor Model

Each synthetic stock i in sector s is built from draws it shares with other stocks and a draw of its own:

$$g_i(t) = \alpha_i + \beta_i \underbrace{g_m(t)}_{\text{market}} + \delta_i \underbrace{f_s(t)}_{\text{sector}} + \psi_i \underbrace{\tilde{\varepsilon}_i(t)}_{\text{own}}$$

α_i, β_i: single index estimates, held fixed. f_s: a model of the sector's average residual, with loading δ_i. $\tilde{\varepsilon}_i$: a model of the stock, centered and scaled by ψ_i so its variance matches the real one on average. Each piece is a jump hidden Markov model.

What the Model Keeps and Gives Up

Composing stocks from shared draws on the same day has consequences:

- Keeps: fat tails and clustered volatility in every draw; stocks in a sector move together through their shared sector draw, all stocks through the market draw.
- Gives up: sector draws are independent, so stocks in different sectors are linked only through the market.
- No copula: a copula reorders each stock's draws across days and would scramble the timing of the volatile periods and jump episodes in them.

An Option: A Drifting β

Holding β_i fixed is a simplification: real β wanders. The model can let $\beta_i(t)$ drift around $\hat{\beta}_i$ as an exponential moving average of random shocks, sized by how much the online estimate of β wanders in the real data.

A trading method that re-estimates β as it trades then faces the same problem in the synthetic market as in the real one.

Example: CHEME-5660-L13a-Example-JumpHMM-SectorFactor-Fall-2026.ipynb

Summary

We built synthetic markets for stress tests, from Markov chains to a jump hidden Markov model of one asset and a sector factor model of many stocks.

Key takeaways

- Markov chains: states, a transition matrix estimated by counting, and a stationary distribution; hidden states emit what we observe.
- Jump hidden Markov model: fat tails from binned Student-t states, and part of the volatility clustering from tuned jump episodes.
- Sector factor model: shared market and sector draws make synthetic stocks move together.

Synthetic markets let us test a method on many paths, not one.